Tadd Mao

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Research Interests

My research spans formal methods, artificial intelligence, and multi-agent reinforcement learning (MARL). Specifically, my research focuses on the faithfulness in autoformalization and also touches scalability in MARL.

Education

University of Toronto Sep. 2023 – Apr. 2027 (expected)

H.B.Sc., Computer Science Major (Focus in Artificial Intelligence) & Mathematics Major *Toronto, ON*

- Cumulative GPA: **3.99 / 4.00**.
- Selected grades: Mathematical Logic (**100%**), Theory of Computation (99%), Abstract Mathematics (97%), Group Theory (97%), Real Analysis (94%), Advanced Calculus (94%), Machine Learning (90%).
- Additional coursework: Deep Learning, Artificial Intelligence, Natural Language Processing, Computational Linguistics, Graph Theory, Principles of Programming Languages.

Papers

Does the Proof Prove It That Way? Faithful Formalization of Elements Proofs

Tadd Mao, Tianjun Zhong, Dhruva Arekar, Yuming Feng, One An, Jiani Huang, Xujie Si, Ziyang Li.

Preprint; under review at AAAI-27. arXiv:2608.15432

- Defined five necessary conditions for how faithfully a formal Lean proof reflects the natural-language proof it claims to represent, and built Pistis, an oracle-guided proof search that generates proofs meeting them.
- On Euclid's *Elements*, formalized Book I in Lean which compiles at least 33× as fast as prior state-of-the-art (SOTA) and is judged more faithful 2.89× (127 blinded human reviews) as often as the prior SOTA formalizations.
- Additionally, I am the first to formalize Books II and III of *Elements*.
- Surfaced 52 gaps in Euclid's proofs and 2 genuine citation errors in the translated text.

Aligning Faithfulness for Theory-Scale Autoformalization

Yuming Feng, **Tadd Mao**, Dhruva Arekar, Tianjun Zhong, Jiani Huang, Xujie Si, Ziyang Li

Under review at ICLR 2027. arXiv:2610.18498

- Found prior faithfulness evaluation methods ignore necessary context to properly evaluate faithfulness.
- Proposed a novel methodology to evaluate faithfulness to resolve this gap, and further establishes *where* the unfaithfulness lies using "links".
- Built FaithAlign, the first theory-scale and context-aware faithfulness benchmark (460 items, 4,982 annotated phrase-level links), that establishes a large gap in LLM-as-judge, as evidenced by GPT-6 Astra reaching only 47.8% defect F1 (a measurement of the ability to detect unfaithfulness).
- My contribution is that I first formalized how we label datapoints, created the UI that allows annotators to label the datapoints, and labelled the majority of the ProofNet section and all of the LeanEuclid/Pistis section of the dataset.

Agent Interest and Capability: When Does Decentralized Value Learning Scale?

Tadd Mao[*], Kirill Utyashev[*], Peter Marbach. [*]Equal contribution.

Under review at AAMAS 2027. PDF

- Defined what it means for decentralized value learning (one value approximation per agent) to be scalable, by requiring that approximation error of the aggregate value function remains bounded as the number of agents grows.
- Proved sufficient conditions of scalability and using this, implemented a concrete class of systems that we empirically show are scalable.

Research Groups

Autoformalization & Faithfulness collaboration May 2026 – present
Xujie Si (U. Toronto) & Ziyang Li (Johns Hopkins) ; with collaborators at U. Pennsylvania *Collaboration*

- Started off as an NSERC USRA project advised by Prof. Xujie Si; led to the paper submissions (above).
- Building off of my faithful proof methodology (AAAI-27) and our novel faithfulness evaluation methodology (ICLR 2027), our current direction is to apply these methods to judge the recent mathematical breakthroughs from OpenAI and Anthropic, with the goal of either validating these mathematical claims, or surfacing issues.

Multi-agent reinforcement learning Aug. 2025 – present
University of Toronto, advised by Prof. Peter Marbach *Toronto, ON*

- Grew from an undergraduate research course into a multi-year project, of which this AAMAS 2027 submission is one output.

Open-Source Contribution

- **Mathlib (Lean 4)** — authored and merged a lemma (`Int.closure_eq_zmultiples`) into Mathlib, the community's core formal-mathematics library; wrote the proof independently and refined it through community review and design discussion. PR #39435

Teaching

Teaching Assistant — MAT309 (Mathematical Logic) Fall 2026
University of Toronto, under Prof. Soheil Homayouni *Toronto, ON*

- Lead 2 of the 3 weekly tutorial sections; also marking and invigilation.

Professional Service

- Sub-reviewer, AAAI 2027.

Honors & Awards

- Nominee, CRA Outstanding Undergraduate Researcher Award (2026) — nominated by the Department of Computer Science at the University of Toronto.
- NSERC Undergraduate Student Research Award (2026).
- Dean's List Scholar (2024, 2025, 2026 — all completed years).
- Later Life Learning Scholarship (2025, 2026).
- Innis College Exceptional Achievement Award (2024).

Skills

Formal methods: Lean 4, Mathlib
Programming: Python, C/C++, Java, R, MATLAB
ML & tools: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas; Slurm, Linux, Git
Natural languages: English, French, Mandarin

Personal

- Winner, Prince George Symphony Orchestra Young Artists Competition — performed the first movement of Rachmaninoff's Piano Concerto No. 2 as piano soloist with the PGSO (video)
- Toured with professional musicians (Yu Yu Liu, Simon Macdonald, Roxanne Dykstra), performing violin in ensemble and piano solo, including a standing ovation for Liszt's *La Campanella*.
- Royal Conservatory of Music: ARCT diploma (piano); Level 10 (violin).

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